Supplementary Materials for "Effects of *IL-6* polymorphisms on individual susceptibility to Allergic Diseases: A Systematic Review and Meta-analysis"

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Supplementary Figure 4. Sensitivity analyses of *IL-6* rs1800795 and allergic diseases risk under the A: Dominant comparison; B: Recessive comparison; C: Allele comparison; D: Heterozygote comparison; E: Homozygote comparison.

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Supplementary Figure 8. Funnel plots of *IL6* rs1800795 and allergic diseases risk under the A: Dominant Model; B: Recessive comparison; C: Allele comparison; D: Heterozygote comparison; E: Homozygote comparison.

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Supplementary Table 1. Search strategies.

Databases	Search items	Result
PubMed	("asthma"[MeSH Terms] OR "asthma"[All Fields] OR "asthmas"[All Fields] OR "asthma s"[All Fields] OR ("asthma"[MeSH Terms] OR "asthma"[All Fields] OR "asthmatic"[All Fields] OR "asthmatics"[All Fields]) OR ("dermatitis, atopic"[MeSH Terms] OR ("dermatitis"[All Fields] AND "atopic"[All Fields]) OR "atopic dermatitis"[All Fields] OR ("atopic"[All Fields] AND "dermatitis"[All Fields])) OR ("rhinitis, allergic"[MeSH Terms] OR ("rhinitis"[All Fields] AND "allergic"[All Fields]) OR "allergic rhinitis"[All Fields] OR ("allergic"[All Fields] AND "rhinitis"[All Fields]))) AND (("interleukin 6"[MeSH Terms] OR "interleukin 6"[All Fields] OR "il 6"[All Fields] OR ("interleukin 6"[MeSH Terms] OR "interleukin 6"[All Fields] OR "interleukin 6"[All Fields])) AND ("polymorphic"[All Fields] OR "polymorphics"[All Fields] OR "polymorphism s"[All Fields] OR "polymorphism, genetic"[MeSH Terms] OR ("polymorphism"[All Fields] AND "genetic"[All Fields]) OR "genetic polymorphism"[All Fields] OR "polymorphism"[All Fields] OR "polymorphisms"[All Fields] OR ("variant"[All Fields] OR "variant s"[All Fields] OR "variants"[All Fields]) OR ("variation"[All Fields] OR "variations"[All Fields]) OR ("mutate"[All Fields] OR "mutated"[All Fields] OR "mutates"[All Fields] OR "mutating"[All Fields] OR "mutation"[MeSH Terms] OR "mutation"[All Fields] OR "mutations"[All Fields] OR "mutation s"[All Fields] OR "mutational"[All Fields] OR "mutator"[All Fields] OR "mutators"[All Fields]) OR ("socioaffect neurosci psychol"[Journal] OR "snp"[All Fields])))	153
Cochrane Library	#1 (IL-6):ti,ab,kw OR (Interleukin-6):ti,ab,kw #2 (polymorphism):ti,ab,kw OR (variant):ti,ab,kw OR (variation):ti,ab,kw OR (mutation):ti,ab,kw OR (SNP):ti,ab,kw #3 (asthma):ti,ab,kw OR (asthmatic):ti,ab,kw OR (atopic dermatitis):ti,ab,kw OR (allergic rhinitis):ti,ab,kw #4 #1 AND #2 AND #3	16
Web of science	#1 TOPIC: (IL-6) OR TOPIC: (Interleukin-6) Databases= WOS, CSCD, KJD, MEDLINE, RSCI, SCIELO Timespan=All years Search language=Auto #2 TOPIC: (polymorphism) OR TOPIC: (variant) OR TOPIC: (variation) OR TOPIC: (mutation) OR TOPIC: (SNP) Databases= WOS, CSCD, KJD, MEDLINE, RSCI, SCIELO Timespan=All years Search language=Auto #3 TOPIC: (asthma) OR TOPIC: (asthmatic) OR TOPIC: (atopic dermatitis) OR TOPIC: (allergic rhinitis) Databases= WOS, CSCD, KJD, MEDLINE, RSCI, SCIELO Timespan=All years Search language=Auto #4 #3 AND #2 AND #1 Databases= WOS, CSCD, KJD, MEDLINE, RSCI, SCIELO Timespan=All years Search language=Auto	325

Supplementary Table 2. Results of pooled ORs in the meta-analysis of the association between *IL-6* gene polymorphisms and overall allergic disease. （After removing the studies of deviation from HWE）.

	Population	Sample size, cases/controls	Dominant comparison			Recessive comparison			Allele comparison			Heterozygote comparison			Homozygote comparison		
			p value	OR (95% CI)	I^2 statistic	p value	OR (95% CI)	I^2 statistic	p value	OR (95% CI)	I^2 statistic	p value	OR (95% CI)	I^2 statistic	p value	OR (95% CI)	I^2 statistic
	Overall	895/1262	0.33	0.74 [0.41, 1.35]	88%	0.32	0.73 [0.39, 1.35]	74%	0.35	0.81 [0.52, 1.26]	89%	0.38	0.79 [0.47, 1.34]	84%	0.24	0.58 [0.24, 1.44]	85%
−174 G/C (rs1800795)	Caucasian	432/765	0.10	0.62 [0.35, 1.10]	77%	0.60	0.81 [0.36, 1.79]	76%	0.09	0.69 [0.46, 1.06]	79%	0.11	0.67 [0.40, 1.10]	66%	0.13	0.49 [0.20, 1.22]	76%
	Asians	373/353	**0.03**	**1.53 [1.05, 2.23]**	**19%**	0.09	1.47 [0.94, 2.31]	NA	**0.02**	**1.39 [1.06, 1.83]**	**12%**	**0.03**	**1.43 [1.03, 1.98]**	**41%**	**0.04**	**1.67 [1.02, 2.74]**	**NA**
	Children	264/250	**P<0.0001**	**0.41 [0.28, 0.61]**	**NA**	**0.0005**	**0.47 [0.31, 0.72]**	**NA**	**P<0.00001**	**0.54 [0.42, 0.69]**	**NA**	**0.006**	**0.48 [0.31, 0.73]**	**NA**	**P<0.00001**	**0.29 [0.18, 0.49]**	**NA**
	Adults	272/533	0.45	0.63 [0.19, 2.09]	90%	0.08	0.38 [0.13, 1.10]	0	0.47	0.73 [0.31, 1.72]	88%	0.53	0.71 [0.25, 2.03]	**89%**	**0.02**	**0.29 [0.10, 0.83]**	**0**

−572 G/C (rs1800796)	Overall	529/515	0.91	1.02 [0.78, 1.32]	0	0.09	1.43 [0.95, 2.17]	0	0.34	1.10 [0.90, 1.35]	0	0.59	0.92 [0.70, 1.23]	0	0.1	1.46 [0.93, 2.27]	0
-597 G/A (rs1800797)	Overall	311/719	0.36	0.88 [0.66, 1.16]	14%	0.80	1.16 [0.36, 3.74]	58%	0.72	0.94 [0.65, 1.35]	55%	0.25	0.84 [0.63, 1.13]	0	0.91	1.08 [0.31, 3.71]	62%
	Caucasian	311/719	0.36	0.88 [0.66, 1.16]	14%	0.80	1.16 [0.36, 3.74]	58%	0.72	0.94 [0.65, 1.35]	55%	0.25	0.84 [0.63, 1.13]	0	0.91	1.08 [0.31, 3.71]	62%
	Adults	74/301	0.08	0.63 [0.37, 1.06]	NA	0.22	0.47 [0.14, 1.59]	NA	0.06	0.66 [0.43, 1.01]	NA	0.16	0.68 [0.39, 1.16]	NA	0.15	0.40 [0.12, 1.38]	NA

Supplementary Table 3. Results of pooled ORs in the meta-analysis of the association between IL-6 gene polymorphisms and each allergic disease. （After removing the studies of deviation from HWE）

Polymorphism and disease	Population	Sample size, cases/controls	Dominant comparison			Recessive comparison			Allele comparison			Heterozygote comparison			Homozygote comparison		
			p value	OR (95% CI)	I^2 statistic	p value	OR (95% CI)	I^2 statistic	p value	OR (95% CI)	I^2 statistic	p value	OR (95% CI)	I^2 statistic	p value	OR (95% CI)	I^2 statistic
−174 G/C (rs1800795) and asthma	Overall	536/783	0.16	0.57 [0.26, 1.25]	**87%**	**0.0001**	**0.46 [0.31, 0.68]**	0	0.12	0.66 [0.39, 1.11]	83%	0.21	0.64 [0.32, 1.28]	**85%**	**P<0.00001**	**0.29 [0.18, 0.46]**	**0**
	Caucasian	338/551	**P<0.00001**	**0.46 [0.33, 0.63]**	**0**	**0.0003**	**0.47 [0.31, 0.70]**	**0**	**P<0.00001**	**0.52 [0.42, 0.63]**	**26%**	**0.0001**	**0.52 [0.37, 0.72]**	**0**	**P<0.00001**	**0.31 [0.19, 0.49]**	**0**
	Children	264/250	**P<0.00001**	**0.41 [0.28, 0.61]**	NA	**0.0005**	**0.47 [0.31, 0.72]**	**NA**	**P<0.00001**	**0.54 [0.42, 0.69]**	**NA**	**0.0006**	0.48 [0.31, 0.73]	**89%**	**P<0.00001**	**0.29 [0.18, 0.49]**	**NA**
	Adults	164/445	**0.02**	**0.67 [0.47, 0.94]**	90%	0.08	0.38 [0.13, 1.10]	0	0.47	0.73 [0.31, 1.72]	88%	**0.02**	**0.67 [0.48, 0.93]**	**89%**	**0.02**	**0.29 [0.10, 0.83]**	**0**

−174 G/C (rs1800795) and allergic rhinitis	Allergic rhinitis	265/265	0.09	1.36 [0.96, 1.94]	NA	0.09	1.47 [0.94, 2.31]	**NA**	**0.03**	**1.31 [1.02, 1.67]**	NA	0.24	1.26 [0.86, 1.84]	**NA**	**0.04**	**1.67 [1.02, 2.74]**	**NA**
−174 G/C (rs1800795) and atopic dermatitis	Atopic dermatitis	94/214	0.71	1.11 [0.65, 1.89]	NA	0.94	0.98 [0.54, 1.79]	NA	0.84	1.04 [0.73, 1.46]	NA	0.67	1.13 [0.64, 1.98]	NA	0.88	1.06 [0.52, 2.13]	NA
-597 G/A (rs1800797) and asthma	Overall	134/441	0.12	0.72 [0.48, 1.09]	0	0.14	0.43 [0.14, 1.32]	0	0.06	0.72 [0.50, 1.01]	0	0.26	0.79 [0.52, 1.20]	0	0.09	0.37 [0.12, 1.17]	0
	Caucasian	134/441	0.12	0.72 [0.48, 1.09]	0	0.14	0.43 [0.14, 1.32]	0	0.06	0.72 [0.50, 1.01]	0	0.26	0.79 [0.52, 1.20]	0	0.09	0.37 [0.12, 1.17]	0

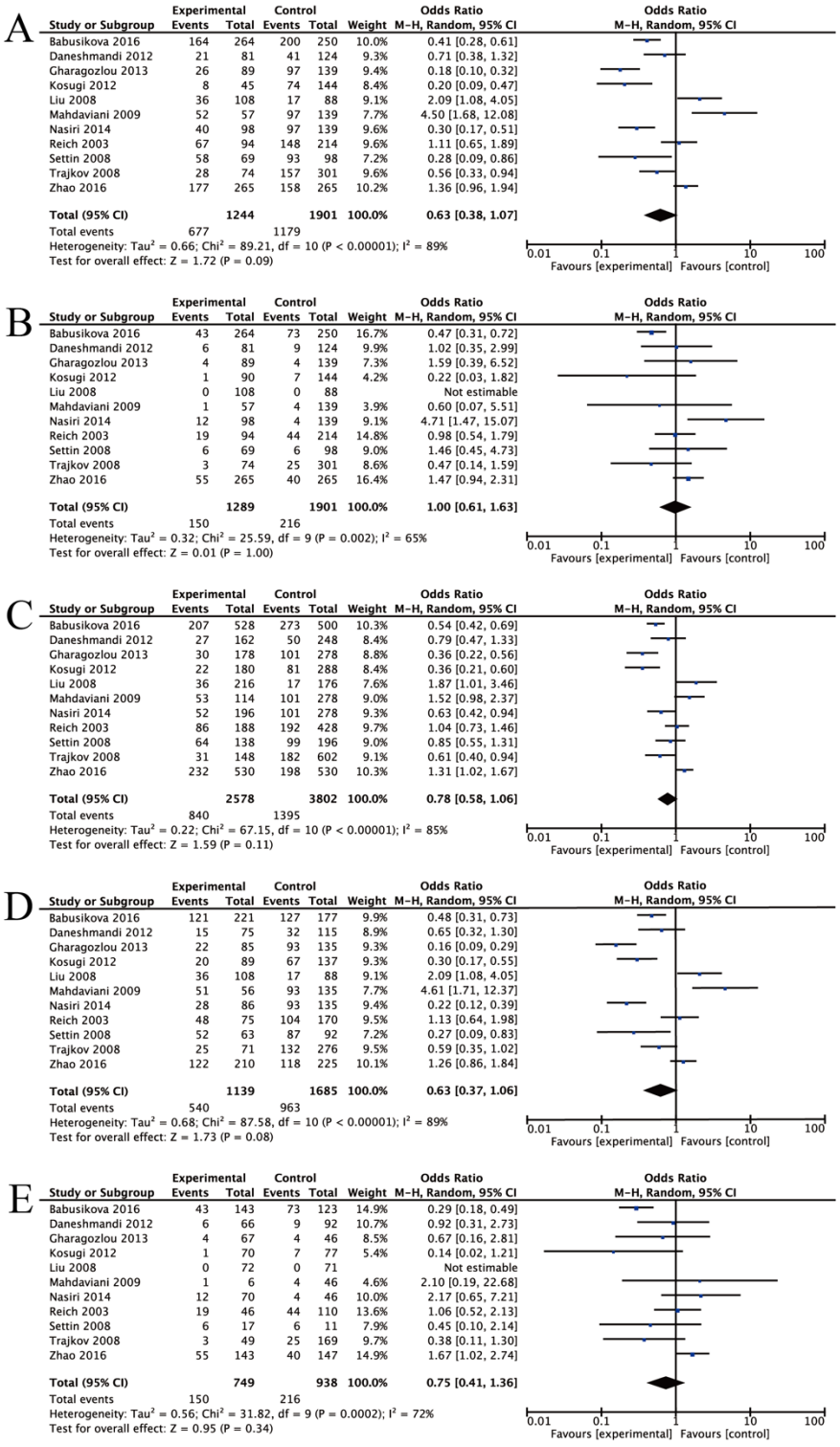
Abbreviations: OR, Odds ratio; CI, Confidence interval.

The values in bold represent there is statistically significant differences between cases and controls.

Supplementary Table 4. *P* value of Egger's tests for *IL6* rs1800795 polymorphism and overall allergic diseases risk under the A: Dominant comparison; B: Recessive comparison; C: Allele comparison; D: Heterozygote comparison; E: Homozygote comparison.

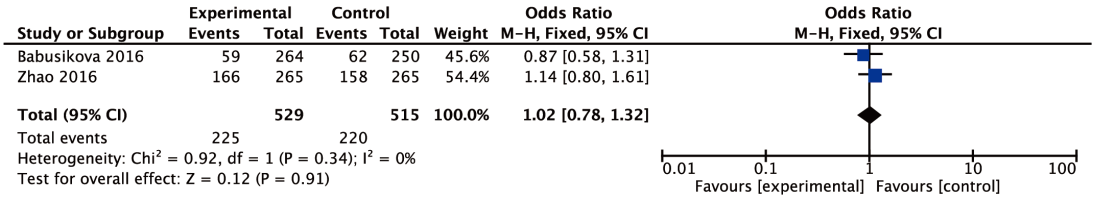
Comparison	Dominant comparison	Recessive comparison	Allele comparison	Heterozygote comparison	Homozygote comparison
P value	0.775	0.723	0.773	0.907	0.886

Supplementary Figure 1. Pooled OR and 95% CI of individual studies and pooled data for the association between *IL-6* rs1800795 polymorphism and overall allergic diseases risk under the A: Dominant comparison; B: Recessive comparison; C: Allele comparison; D: Heterozygote comparison; E: Homozygote comparison.



Supplementary Figure 2. Pooled OR and 95% CI of individual studies and pooled data for the association between *IL-6* rs1800796 polymorphism and overall allergic diseases risk under the A: Dominant comparison; B: Recessive comparison; C: Allele comparison; D: Heterozygote comparison; E: Homozygote comparison.

A



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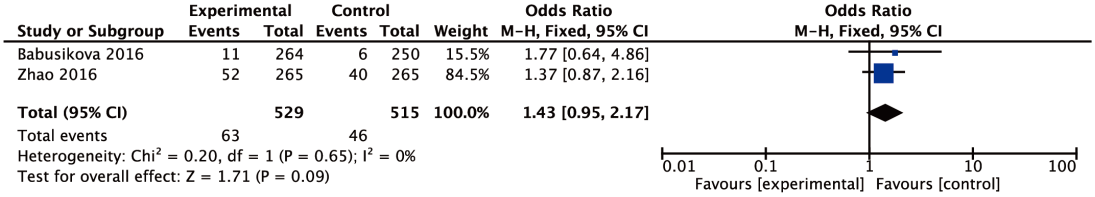

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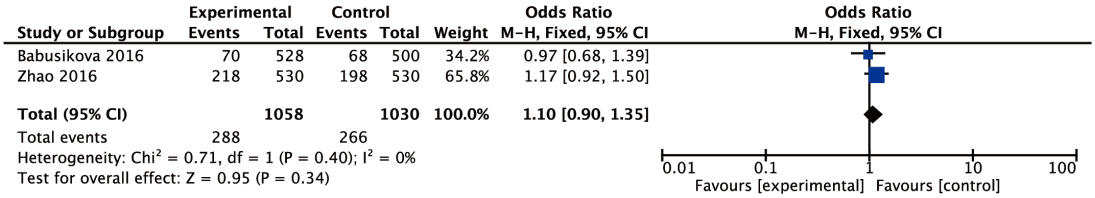

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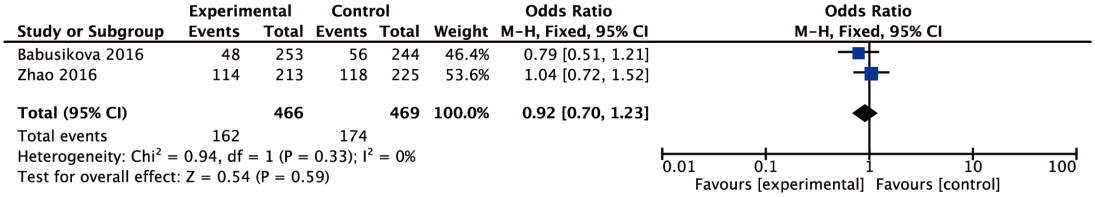

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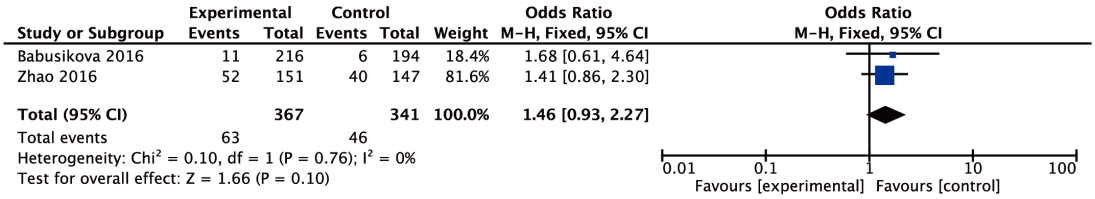

Supplementary Figure 3. Pooled OR and 95% CI of individual studies and pooled data for the association between *IL-6* rs1800797 polymorphism and overall allergic diseases risk under the A: Dominant comparison; B: Recessive comparison; C: Allele comparison; D: Heterozygote comparison; E: Homozygote comparison.

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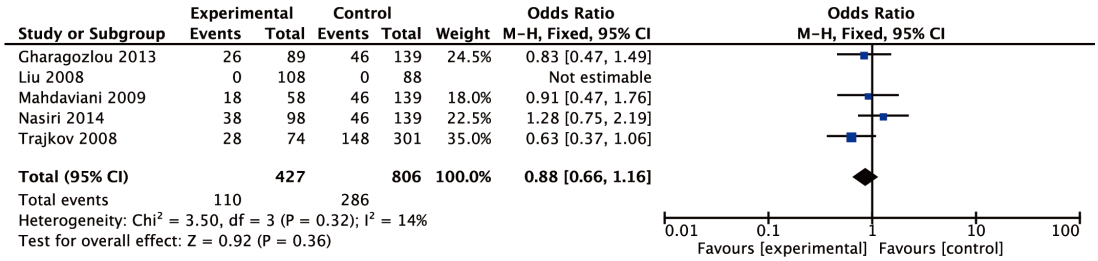

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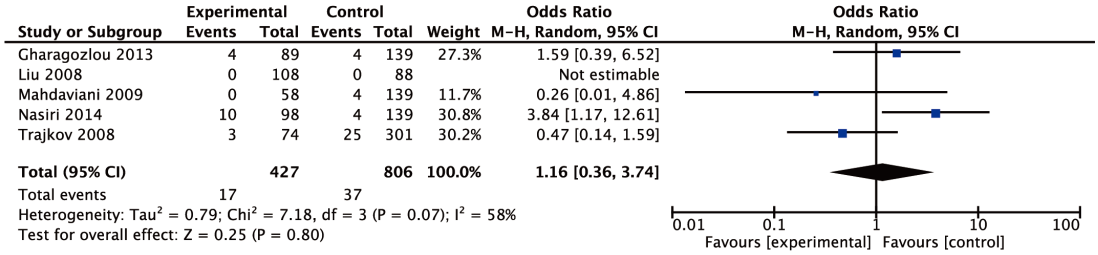

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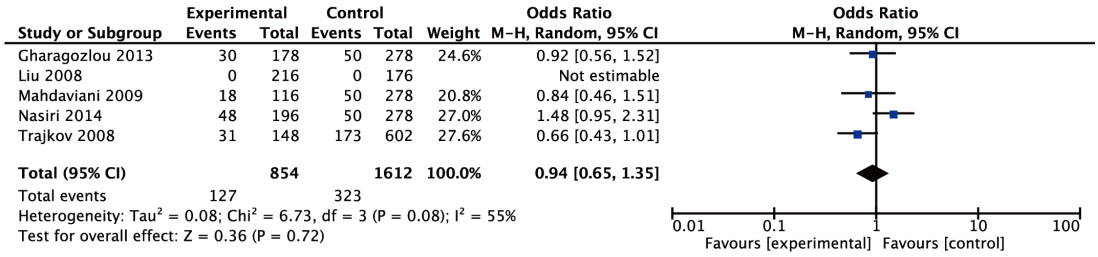

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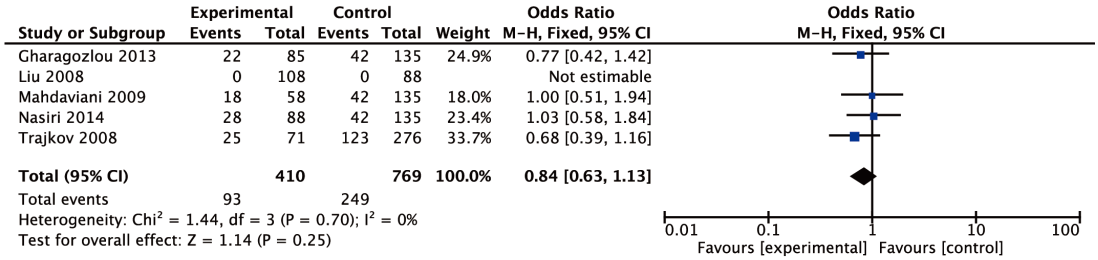

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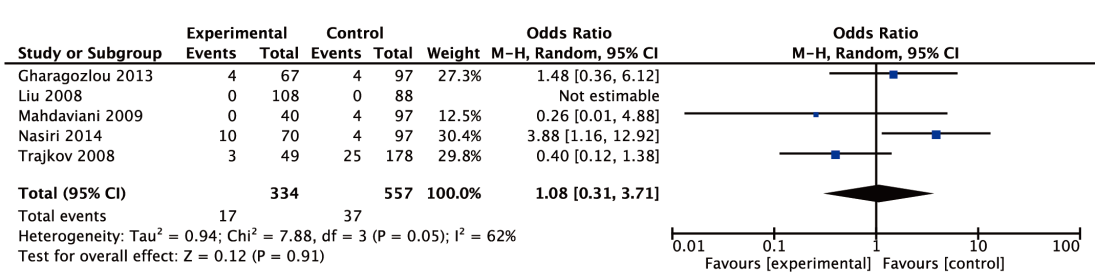
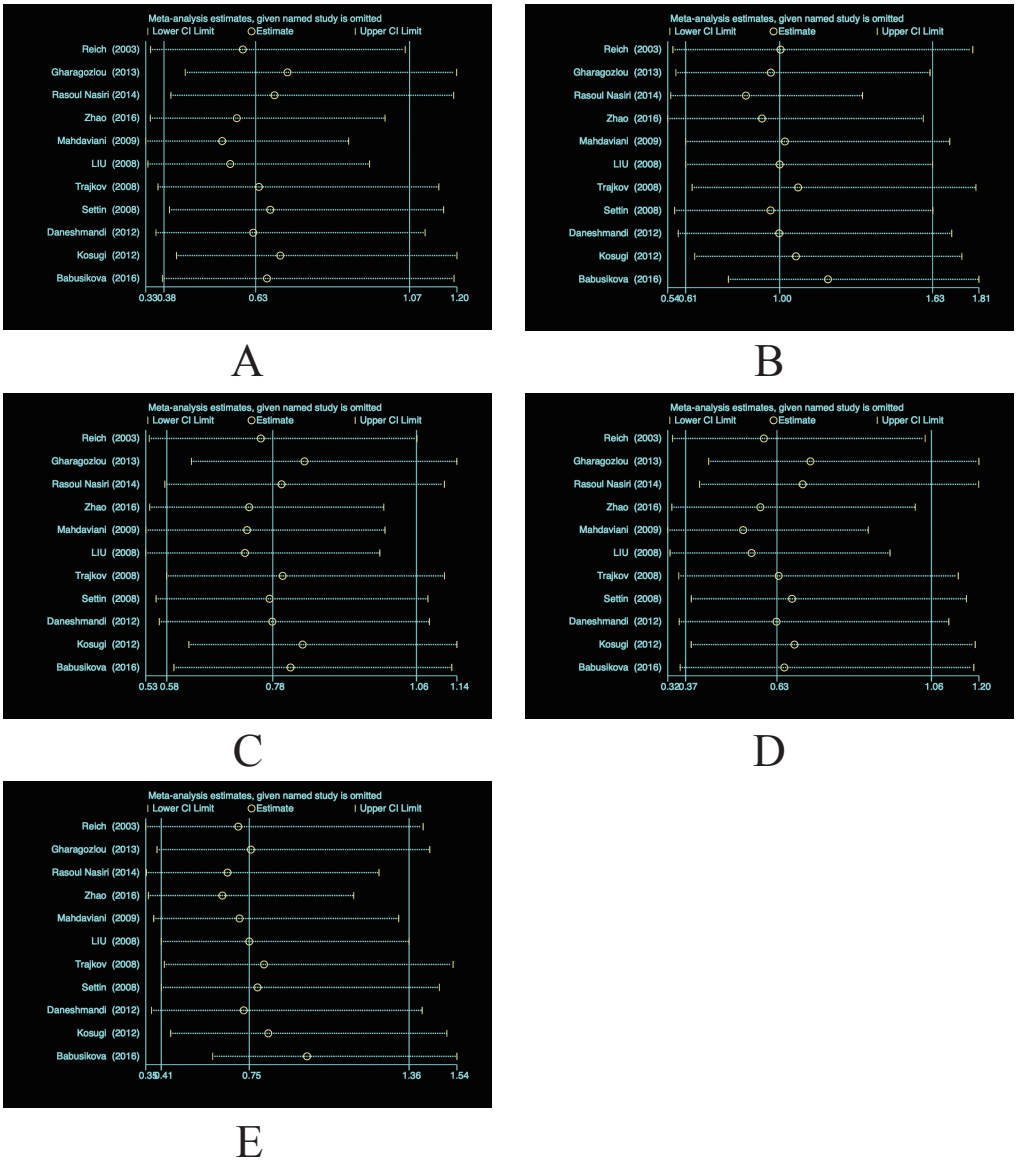
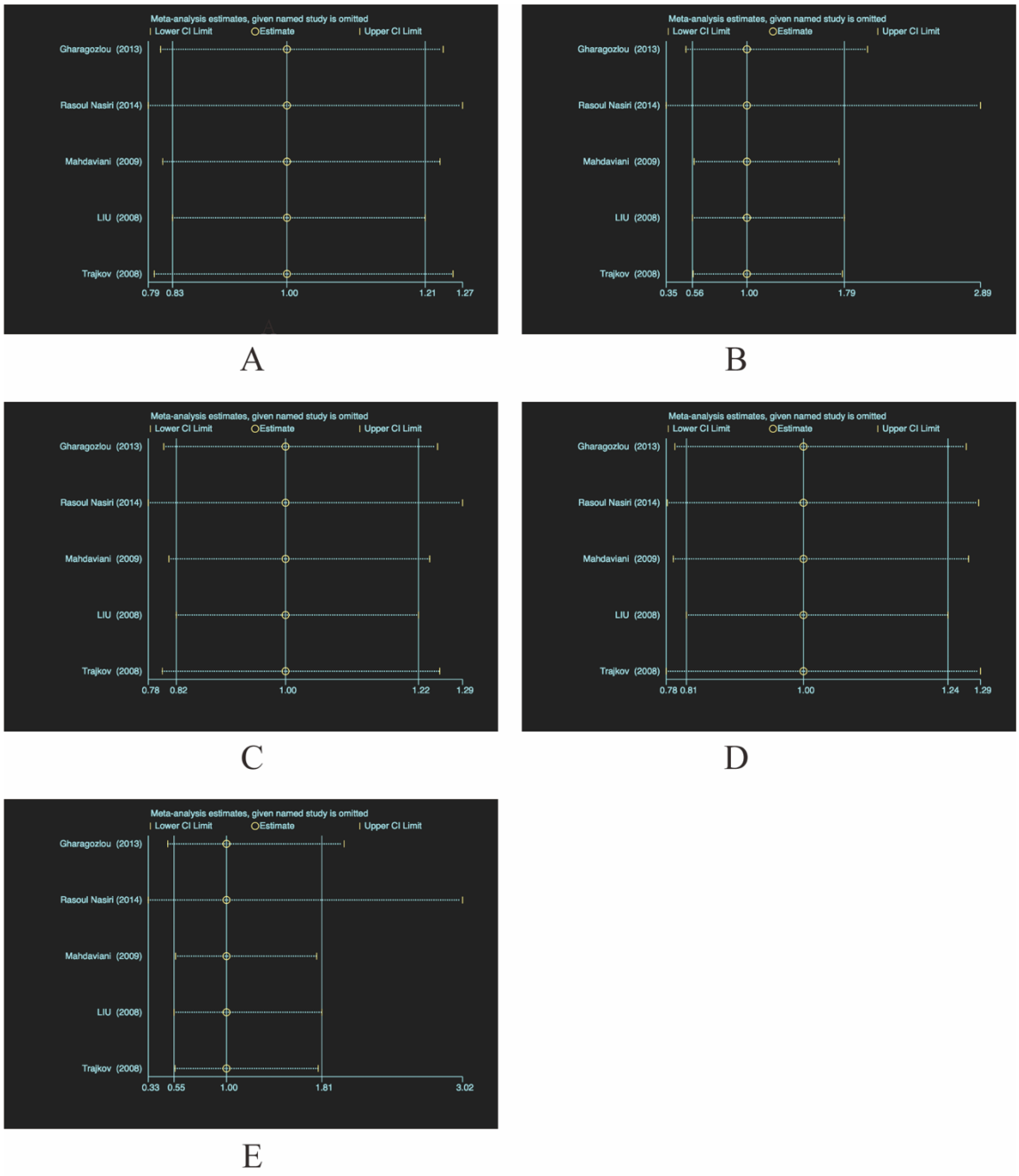


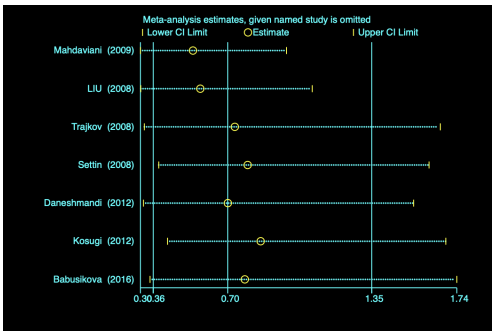
Supplementary Figure 4. Sensitivity analyses of *IL-6* rs1800795 and overall allergic diseases risk under the A: Dominant comparison; B: Recessive comparison; C: Allele comparison; D: Heterozygote comparison; E: Homozygote comparison.



Supplementary Figure 5. Sensitivity analyses of *IL-6* rs1800797 and overall allergic diseases risk under the A: Dominant comparison; B: Recessive comparison; C: Allele comparison; D: Heterozygote comparison; E: Homozygote comparison.



Supplementary Figure 6. Sensitivity analyses of *IL-6* rs1800795 and asthma risk under the A: Dominant comparison; B: Recessive comparison; C: Allele comparison; D: Heterozygote comparison; E: Homozygote comparison.



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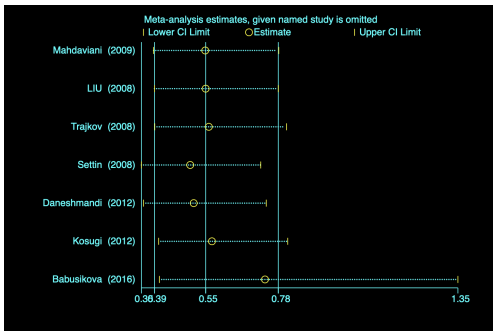

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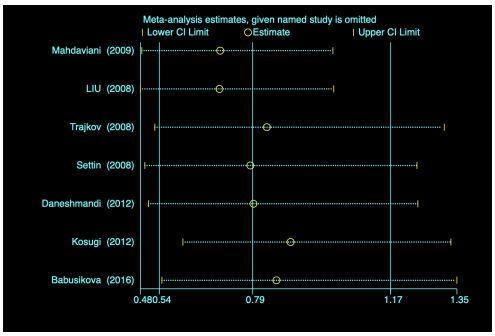

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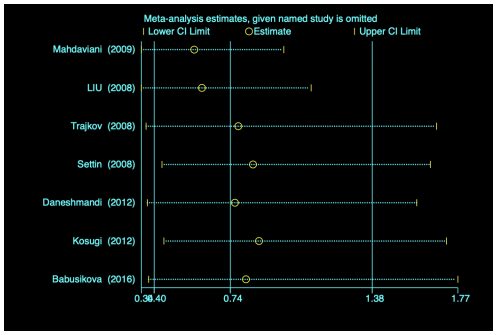

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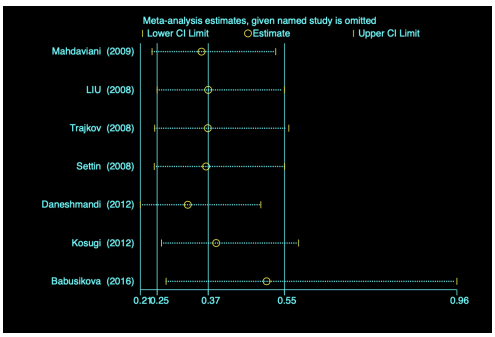

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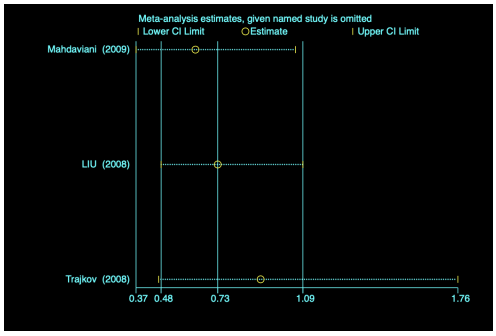
Supplementary Figure 7. Sensitivity analyses of *IL-6* rs1800797 and asthma risk under the A: Dominant comparison; B: Recessive comparison; C: Allele comparison; D: Heterozygote comparison; E: Homozygote comparison.



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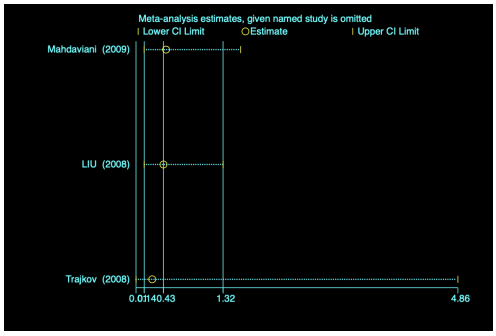

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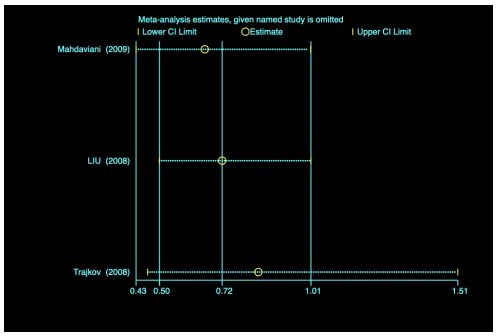

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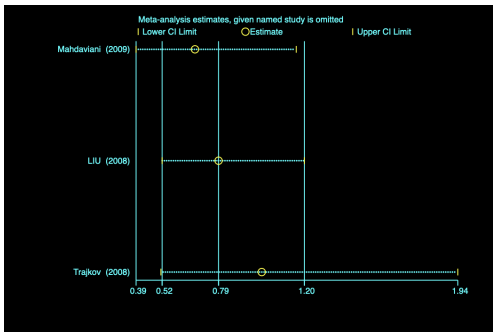

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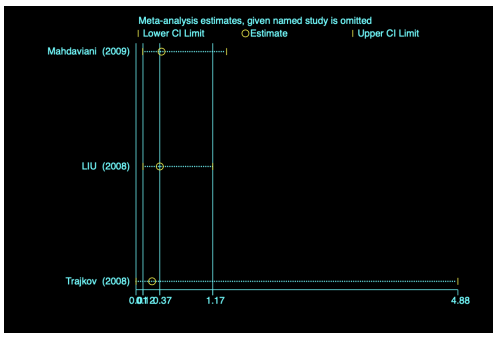

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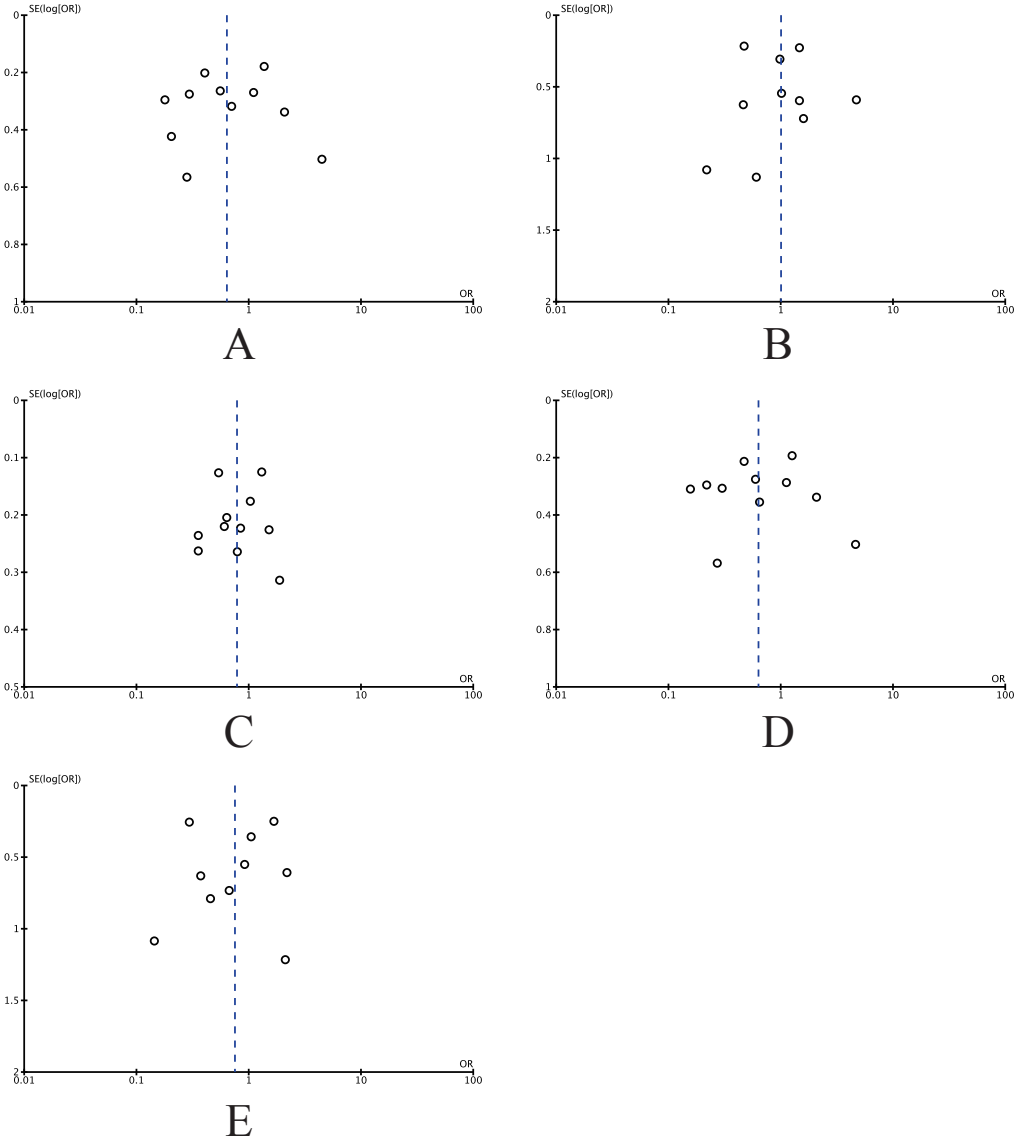
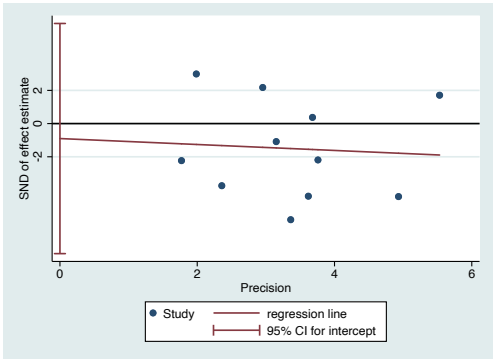
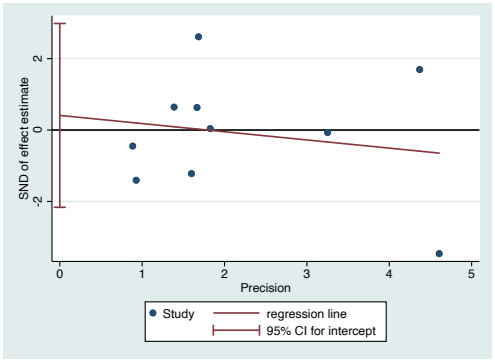
Supplementary Figure 8. Funnel plots of *IL6* rs1800795 and overall allergic diseases risk under the A: Dominant comparison; B: Recessive comparison; C: Allele comparison; D: Heterozygote comparison; E: Homozygote comparison.



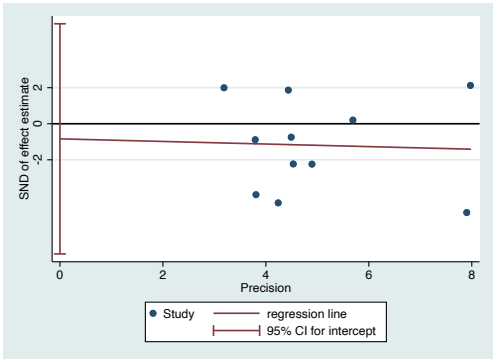
Supplementary Figure 9. Egger's tests for *IL-6* rs1800795 and overall allergic diseases risk under the A: Dominant comparison; B: Recessive comparison; C: Allele comparison; D: Heterozygote comparison; E: Homozygote comparison.



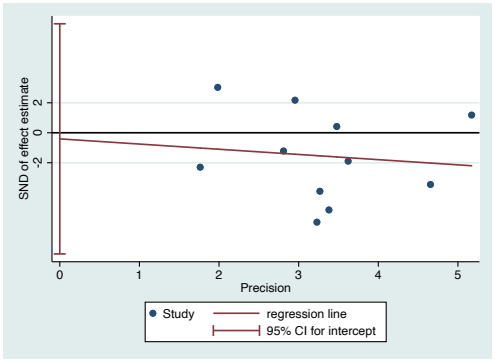
A



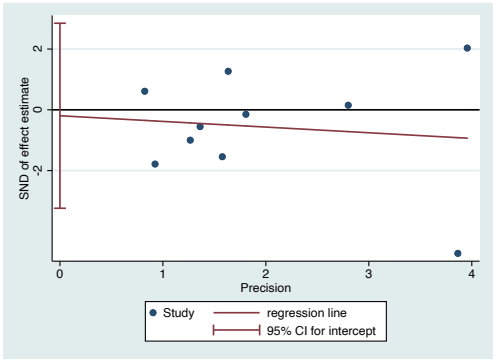
B



C



D



E

Supplementary Figure 10. The process of Egger's tests for *IL-6* rs1800795 and overall allergic diseases risk under the A: Dominant comparison; B: Recessive comparison; C: Allele comparison; D: Heterozygote comparison; E: Homozygote comparison.

```
. metabias _ES _seES, egger graph

Note: data input format theta se_theta assumed

Egger's test for small-study effects:
Regress standard normal deviate of intervention
  effect estimate against its standard error

.
Number of studies = 11                                  Root MSE      =   3.133

------------------------------------------------------------------------------
     Std_Eff |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
       slope |   -.1791953   .8643343    -0.21   0.840    -2.134455    1.776065
        bias |   -.9016096   3.064687    -0.29   0.775    -7.834413    6.031194
------------------------------------------------------------------------------

Test of H0: no small-study effects          P = 0.775
```

A

```
. metabias _ES _seES, egger graph

Note: data input format theta se_theta assumed

Egger's test for small-study effects:
Regress standard normal deviate of intervention
  effect estimate against its standard error

.
Number of studies = 10                                  Root MSE      =   1.774

------------------------------------------------------------------------------
     Std_Eff |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
       slope |   -.229227   .4345932    -0.53   0.612    -1.231401    .7729467
        bias |   .4102785   1.116147     0.37   0.723    -2.163562    2.984119
------------------------------------------------------------------------------

Test of H0: no small-study effects          P = 0.723
```

B

```
. metabias _ES _seES, egger graph

Note: data input format theta se_theta assumed

Egger's test for small-study effects:
Regress standard normal deviate of intervention
  effect estimate against its standard error

.
Number of studies = 11                                  Root MSE      =   2.718

------------------------------------------------------------------------------
     Std_Eff |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
       slope |   -.0716735   .5404149    -0.13   0.897    -1.294177    1.15083
        bias |   -.8379512   2.821281    -0.30   0.773    -7.220133    5.54423
------------------------------------------------------------------------------

Test of H0: no small-study effects          P = 0.773
```

C

```
. metabias _ES _seES, egger graph

Note: data input format theta se_theta assumed

Egger's test for small-study effects:
Regress standard normal deviate of intervention
  effect estimate against its standard error

.
Number of studies = 11                                  Root MSE      =   3.116

------------------------------------------------------------------------------
     Std_Eff |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
       slope |   -.3467979   .9881356    -0.35   0.734    -2.582116    1.88852
        bias |   -.4067153   3.394815    -0.12   0.907    -8.08632     7.27289
------------------------------------------------------------------------------

Test of H0: no small-study effects          P = 0.907
```

D

```
. metabias _ES _seES, egger graph

Note: data input format theta se_theta assumed

Egger's test for small-study effects:
Regress standard normal deviate of intervention
  effect estimate against its standard error

.
Number of studies = 10                                  Root MSE      =   1.991

------------------------------------------------------------------------------
     Std_Eff |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
       slope |   -.1865779   .5799541    -0.32   0.756    -1.523954    1.150799
        bias |   -.1951114   1.320405    -0.15   0.886    -3.239971    2.849748
------------------------------------------------------------------------------

Test of H0: no small-study effects          P = 0.886
```

E